

**THE IMPACT OF CYBERWARFARE ON WAR AND
PRIVACY OF DEPRIVED NATION-STATE**

(A CASE STUDY OF EVENTS BETWEEN 2015 - 2025)

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IHERIS/TG/2022/009/IRS

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DECLARATION OF ORIGINALITY

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This project report is submitted as part of the requirement for the Bachelor of Science in International Relations at the Institut Des Hautes Etudes De Relations Internationales & Strategiques Togo. I confirm that the work contained in this B.Sc. project report has been solely composed by myself, except where indicated in the text, and has not been accepted in any previous application/award for a degree in any institution. The report may be freely copied and distributed provided the source is acknowledged.

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CERTIFICATION OF APPROVAL

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Meets the requirements governing the award of the degree of **BACHELOR OF SCIENCE** in **INTERNATIONAL RELATIONS** and is hereby approved for its contribution to knowledge and literary presentation.

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DEDICATION

This research project is, above all, dedicated to the bold and tireless individuals standing guard at the front lines of our digital world.

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ABSTRACT

This study delves into how cyberwarfare shapes international conflicts and threatens the privacy of nation-states. It focuses on notable incidents such as the Stuxnet attack, the 2016 U.S. election interference, and the OPM data breach. The research aims to explore the strategic impacts of cyberwarfare, examine the ways it compromises privacy, and assess whether current cybersecurity policies are up to the task. Central questions revolve around shifts in national security, privacy risks, legal protections, and international cooperation, with hypotheses testing these areas.

Adopting a qualitative case study approach, the study draws on secondary sources including government reports, academic papers, and media coverage. Thematic analysis helps uncover the nature, techniques, and broader implications of cyber-attacks, along with their effects on privacy.

The focus lies on nation-states affected by cyberwarfare from 2007 through 2025. Results highlight a clear erosion of privacy rights amid strategic cyber operations, while current legal frameworks fall short of fully addressing these threats. The study wraps up with calls for stronger international cooperation, improved legal standards, and ethical cyber policies to better safeguard national security and individual privacy in today's digital age.

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NOMENCLATURE/LIST OF ABBREVIATIONS OR ACRONYMS

AI	Artificial Intelligence
AU	African Union
CIA	Central Intelligence Agency
CISA	Cybersecurity and Infrastructure Security Agency
DDoS	Distributed Denial-of-Service
DNC	Democratic National Committee
DNI	Director of National Intelligence
ECOWAS	Economic Community of West African States
GDPR	General Data Protection Regulation
GERD	Grand Ethiopian Renaissance Dam
GRU	Main Directorate of the General Staff (Russia's Military Intelligence)
ICRC	International Committee of the Red Cross
IDS	Intrusion Detection System
IP	Internet Protocol
IRA	Internet Research Agency
IT	Information Technology

KGB	Komitet Gosudarstvennoy Bezopasnosti (Soviet Union)
MFA	Multi-Factor Authentication
NATO	North Atlantic Treaty Organization
NSA	National Security Agency
OPM	Office of Personnel Management
OSINT	Open Source Intelligence
R&D	Research and Development
SOC	Security Operations Center
UN	United Nations
UNHRC	United Nations Human Rights Council
URL	Uniform Resource Locator
U.S.	United States
VPN	Virtual Private Network
VTC	Video Teleconference
WiFi	Wireless Fidelity
XML	Extensible Markup Language

CHAPTER 1 INTRODUCTION

In today's fast-paced digital age, where technological innovation intersects with rising geopolitical tension, cyberwarfare has emerged as a powerful and increasingly pivotal aspect of international conflict. It involves the calculated use of cyber-attacks by states to disrupt, damage, or destabilize critical infrastructure, economic systems, and national security mechanisms of rival nations. Unlike traditional forms of warfare, cyberwarfare unfolds in the digital realm, often obscuring distinctions between times of peace and conflict, and complicating efforts to attribute attacks or respond proportionately.

A broad review of the literature reveals a spectrum of interpretations regarding cyberwarfare's impact. Rid (2013) contends that cyber capabilities have redefined conflict by enabling asymmetric strategies, allowing smaller states and even non-state actors to effectively challenge global powers. Conversely, Libicki (2012) draws attention to the inherent ambiguity of cyber operations, which makes deterrence and the development of international norms more difficult.

A particularly pressing yet often underexplored consequence of cyberwarfare is the erosion of privacy. Schneier (2015) points out that cyber espionage and state surveillance not only breach personal data but also undermine democratic institutions and public confidence. Deibert (2013) similarly emphasizes the serious ethical and legal tensions that arise from mass surveillance and intrusive digital tactics, prompting critical debate over the balance between protecting national security and preserving civil liberties.

This study unpacks the dual impacts of cyberwarfare, its influence on global conflict dynamics and the implications for privacy rights within and across states. Through a careful synthesis of scholarly perspectives, the research aims to clarify the complexities between cyber operations, state behavior, and individual freedoms, while underscoring the need for stronger international frameworks to manage these emerging challenges.

The ultimate goal is to offer a well-rounded understanding of cyberwarfare's layered realities and to suggest actionable, balanced solutions that uphold both national interests and fundamental human rights in our increasingly interconnected world.

1.1 BACKGROUND OF THE STUDY

The Cyberwarfare has become a defining factor in modern international relations, shaped by rapid technological progress and the rising strategic value of cyberspace. In the past few years, academic interest in this field has intensified, reflecting a deeper and more nuanced understanding of the threats posed by digital conflict.

Notable cyber incidents, such as the 2020 SolarWinds breach and the 2021 Colonial Pipeline attack, have demonstrated the increasing sophistication and disruptive power of state-linked cyber operations. Analysts like Healey (2020) and Valeriano and Jensen (2019) have emphasized the growing frequency and impact of these events, noting their far-reaching consequences on national security, global economies, and international diplomacy.

Recent scholarship such as the work of Singer and Brooking (2018) and Buchanan (2020) highlights the unique attributes of cyberwarfare, including its ambiguity, potential for rapid escalation, and exploitation of systemic technological weaknesses. These factors, they argue, make traditional security responses insufficient, calling for adaptive and forward-looking defence strategies.

Alongside national defence concerns, the erosion of privacy through cyber operations has drawn increasing scrutiny. Researchers like Zuboff (2019) and Ferguson (2021) warn that widespread surveillance and cyber espionage are undermining individual rights and democratic norms. These intrusions often extend beyond direct victims, affecting entire populations and weakening trust in digital systems and public institutions.

This study investigates these evolving dynamics, critically examining how cyberwarfare influences both global security structures and the protection of personal

privacy. Drawing from contemporary literature and real-world cyber events, it aims to deliver a timely, comprehensive perspective on the intersection of cybersecurity, international conflict, and human rights. The research ultimately advocates for resilient, ethical, and responsive cybersecurity strategies that uphold both state security and civil liberties in the digital era.

1.2 STATEMENT OF THE STUDY

As international relations and digital security continue to evolve rapidly, cyberwarfare has emerged as a powerful and increasingly common threat, disrupting national stability, economic systems, and personal privacy on a global scale. Despite significant investments in cybersecurity and the rollout of various defensive protocols, many nation-states remain vulnerable to advanced cyber-attacks including espionage, infrastructure sabotage, and widespread breaches of privacy.

A key concern is the insufficient preparedness of governments to counter these evolving threats. Major incidents like the 2020 SolarWinds breach and the 2021 Colonial Pipeline attack have exposed serious flaws in existing security systems and highlighted the ineffectiveness of current international norms in addressing such threats.

Compounding the issue is the growing infringement on privacy rights. Under the guise of national security, state actors increasingly engage in mass surveillance and intrusive cyber espionage, raising ethical concerns and weakening public trust in digital platforms and democratic institutions. These practices challenge foundational principles such as transparency, accountability, and civil liberty.

This study addresses the urgent need to examine how cyberwarfare simultaneously undermines national security and erodes privacy. By identifying the limitations of present-day cybersecurity strategies and international policies, and analysing relevant case studies and literature, the research aims to develop practical, balanced, and ethically sound recommendations that reinforce both state defence and individual rights in our interconnected digital world.

1.3 AIM AND OBJECTIVES

To critically assess how cyberwarfare influences national security strategies and personal privacy, and to evaluate current legal and policy responses with the goal of proposing practical, ethical solutions.

This research paper will intricately address the following objectives.

1. To analyze how cyberwarfare events from 2015 to 2025 have altered national security policies and strategic planning in nation-states.
2. To examine the scope and nature of privacy violations tied to state-sponsored cyber operations between 2015 and 2025.
3. To assess the strengths and weaknesses in international laws and national cybersecurity policies in addressing cyberwarfare and privacy issues from 2015 to 2025.
4. To identify and recommend solutions to the challenges and opportunities for global cooperation in combating cyber threats and protecting privacy rights, as observed in 2015 to 2025.

By addressing these objectives, the study intends to offer practical insights and informed recommendations to policymakers, security professionals, and academics enhancing both the understanding and management of cyberwarfare's dual impact on national defense and individual rights.

1.4 RESEARCH QUESTIONS/HYPOTHESIS

1.4.1 RESEARCH QUESTIONS

This study is guided by the following core research questions:

1. How have cyberwarfare incidents shaped changes in national security policies and strategies?
2. In what ways do cyberwarfare operations affect the privacy rights of individuals and organizations?

3. How effective are current international legal frameworks and national cybersecurity policies in addressing the challenges of cyberwarfare and privacy?
4. What obstacles and drivers influence international cooperation on cyberwarfare and privacy protection?

1.4.2 RESEARCH HYPOTHESES

The study advances the following null hypotheses:

- I. **H₀₁:** Recent cyberwarfare incidents have not significantly influenced national security policies or strategic planning.
- II. **H₀₂:** Cyberwarfare operations do not significantly impact individual or organizational privacy rights.
- III. **H₀₃:** Existing international legal frameworks and national cybersecurity policies are sufficient to effectively manage cyberwarfare and privacy challenges.
- IV. **H₀₄:** There are no significant barriers or facilitators affecting international cooperation on cyberwarfare and privacy protection.

1.5 SIGNIFICANCE OF THE STUDY

This study is of notable importance both academically and practically, as cyberwarfare continues to shape global security dynamics and directly impact individual privacy rights. With digital conflicts becoming more frequent and complex, there is a pressing need for fresh, comprehensive research that reflects these rapidly evolving challenges and their wider societal effects.

Academically, the research adds value by offering a timely, critical analysis of cyberwarfare grounded in recent events and contemporary scholarship. It addresses a clear gap in existing studies by integrating national security concerns with the pressing issue of privacy erosion thereby enriching academic discourse on how cyber conflict redefines traditional war paradigms and civil liberties.

Practically, the study provides meaningful insights for policymakers, cybersecurity professionals, and regulatory institutions. By identifying flaws in current strategies and legal mechanisms, it advocates for more adaptive, ethically responsible approaches that can better align security needs with the protection of individual rights.

Beyond the policy and academic realms, this research also aims to inform the public about the privacy risks associated with cyberwarfare. In raising awareness, it encourages more informed civic engagement and supports collective efforts to strengthen privacy safeguards in an increasingly digital society.

In essence, the significance of this study lies in its capacity to inform future policies, refine institutional responses to cyber threats, and promote a secure, rights-respecting digital environment at both national and global levels.

1.6 SCOPE AND DELIMITATIONS OF THE STUDY

This research is specifically delimited to the analysis of major cyberwarfare events and their impact on national security and privacy between the year 2015 and 2025. The study draws its primary case study from the following selected events:

Case Study Focus:

- I. **Ukraine-Russia Cyberwar (2015–2023):** Including NotPetya (2017), BlackEnergy (2015–2016), Sandworm, and the 2022 invasion cyber-attacks targeting infrastructure and civilian data in Ukraine.
- II. **SolarWinds Hack (2020):** Affecting U.S. government agencies and global enterprises, attributed to Russian state actors, with substantial espionage and privacy implications.
- III. **Colonial Pipeline Ransomware Attack (2021):** Showing the direct link between cyberwarfare/criminals and national infrastructure.

- iv. **U.S. 2016 & 2020 Election Interference:** Focused on Russian and Chinese-backed cyber and disinformation operations targeting democratic processes and privacy.
- v. **Bybit \$1.5B Hack (2025):** A large-scale cryptocurrency attack with state-backed involvement, reflecting economic impact and privacy breaches.
- vi. **Thiopia–Egypt Cyber Attacks On The Grand Ethiopian Renaissance Dam (Gerd) – 2020:** The Grand Ethiopian Renaissance Dam (GERD) has stood at the centre of escalating geopolitical tension between Ethiopia and downstream nations, particularly Egypt.

Only events with credible documentation and state attribution from 2015 to 2025 are analyzed in details. Earlier events (pre-2015) are referenced for context but not as primary evidence.

This research has the following delimitations:

- I. Excludes purely criminal or non-state cyber-attacks not linked to nation-states.
- II. Does not cover technical development of malware or detailed cryptographic analysis.
- III. Focuses on strategic, legal, and privacy impacts as related to international relations.

1.7 DEFINITION OF TERMS

Attribution: The process of identifying the responsible party behind a cyberattack, often complicated by technical obfuscation and geopolitical interests.

Cyberwarfare: Hostile activities conducted via digital networks by states or state-backed actors to achieve military, political, or strategic objectives.

Disinformation: False or misleading information deliberately spread to deceive or influence public perception, often amplified during cyber operations.

DDoS: cyberattack that overwhelms a target system, service, or network with excessive traffic, rendering it unavailable to legitimate users.

Espionage: The unauthorized access and theft of sensitive information for strategic or political advantage, commonly perpetrated in cyberspace.

Infrastructure: Critical systems and assets such as energy, water, transportation, and communication networks that are essential for a nation's functioning.

Malware: Malicious software designed to infiltrate, damage, or gain unauthorized access to computer systems.

Ransomware: A type of malware that encrypts a victim's data or system, demanding payment for its release or decryption.

Sovereignty: The authority of a state to govern itself and control its digital and physical territory, including cyberspace.

Surveillance: Monitoring of digital or physical activities, often by governments or organizations, for security or intelligence purposes.

CHAPTER 2 LITERATURE REVIEW

In recent years, academic interest in cyberwarfare has grown significantly, mirroring its rising impact on global security and digital policy. This literature review positions the current study within that expanding body of research by examining key theories, analytical frameworks, and empirical studies related to the strategic use of cyberwarfare and its broader implications for privacy and international relations.

Drawing on peer-reviewed articles, policy reports, and case analyses published over the past five years, the review identifies core themes, emerging debates, and notable gaps in the existing scholarship. Its objective is to establish a strong conceptual and analytical basis for understanding how cyberwarfare has developed, how states employ it, and how it intersects with the sensitive and evolving issues of personal privacy and digital rights.

2.1 CONCEPTUAL FRAMEWORK

2.1.1 Historical And Contemporary Role Of Cyberwarfare

Though the roots of cyberwarfare trace back to the Cold War, its strategic prominence surged in the early 2000s. Early operations like Moonlight Maze (a mid-1990s U.S. breach traced to Russia) and Titan Rain (a 2003 intrusion attributed to China) signalled a new era of digital espionage. But it was the 2007 DDoS attacks on Estonia, during a political clash with Russia, that marked a watershed moment. These coordinated digital assaults crippled banks, media outlets, and government sites, revealing just how vulnerable modern infrastructure is to cyber disruption even in technologically advanced nations.

The 2008 Russia–Georgia conflict introduced a more integrated approach. As Russian troops advanced on the ground, cyberattacks defaced Georgian websites and overwhelmed state communication systems. While not militarily decisive, these digital offensives compounded confusion and delayed response efforts. The incident illustrated how even limited cyber operations can amplify the chaos of kinetic warfare.

Perhaps the most famous cyber weapon remains Stuxnet (2010)—a sophisticated worm, reportedly crafted by the U.S. and Israel, which sabotaged Iran’s nuclear centrifuges. By exploiting zero-day vulnerabilities and breaching isolated systems, Stuxnet physically damaged hardware using code alone. Its success demonstrated cyberwarfare’s potential to inflict kinetic damage, prompting what many see as the beginning of a global cyber arms race.

Since then, digital tactics have featured in conflicts across the globe. During the Arab Spring, governments like Syria and Iran reportedly deployed spyware to surveil dissidents. The Shamoon virus (2012) wiped data from Saudi Aramco systems, while the BlackEnergy malware targeted Ukrainian infrastructure during the 2014–15 conflict with Russia. Meanwhile, North Korea’s hacks from the Sony Pictures breach (2014) to WannaCry ransomware (2017) demonstrated how isolated regimes can use cyberattacks for coercion and fundraising, blurring the line between state action and cybercrime.

The 2016 U.S. election interference, orchestrated by Russian-linked actors, shifted focus from technical disruption to information warfare. Through phishing, bot networks, and coordinated disinformation, these groups eroded public trust and political stability. Analysts now argue that combining hacking with misinformation may be among the most effective cyber tactics in political conflicts.

Most recently, the 2022 Russia–Ukraine war has seen cyberwarfare used in tandem with military force. Russia launched operations like WhisperGate malware and a major Viasat satellite modem hack, but Ukraine, hardened by years of prior attacks, mounted an effective digital defence. While cyber tools played a strategic role, they did not deliver decisive victories underscoring both the potential and the limitations of cyberwarfare as a standalone strategy.

Across these cases, one trend is clear: cyber operations are now an embedded feature of modern conflict, used by states and non-state actors alike to disrupt, deceive, and dominate across both digital and physical domains.

2.1.2 Concept Of Cyberwarfare

Cyberwarfare is a complex and evolving domain, often lacking a universally accepted definition but generally understood as the use of digital tools such as cyber-attacks and malware by one state against another to achieve strategic goals. Unlike traditional warfare, which relies on physical combat, cyberwarfare occurs in the digital realm and employs tactics like malicious software, denial-of-service (DoS) attacks, and espionage. Sampanis et al. (2023) argue that these operations are specifically designed to disrupt or damage critical infrastructure, steal sensitive data, and undermine national security.

The challenges of regulating cyberwarfare are highlighted by Jantzen (2017), who notes that while the physical boundaries of war are typically clear, cyberspace presents a less defined battlefield. This ambiguity poses significant legal and governance challenges, complicating the application of international law to cyber conflicts.

Recent scholarship, such as Gervais's (2022) analysis, has increasingly recognized cyberspace as the fifth domain of warfare, alongside land, sea, air, and space. The anonymity of cyberspace, combined with its global reach, enables states to pursue strategic interests without direct military engagement. Sampanis et al. (2023) emphasize the broad spectrum of cyberwarfare tactics, from espionage (stealing sensitive data) to propaganda (disinformation campaigns), each of which can have profound effects on both military operations and civilian life.

2.1.3 Categories Of Cyberwarfare: Espionage, Sabotage, And Propaganda

Cyberwarfare encompasses various categories of actions, often overlapping to create both military and civilian impacts.

Espionage involves unauthorized access to sensitive information, such as government documents or corporate secrets, often to gain a strategic advantage. Gervais (2022) highlights data theft as one of the most common forms of

cyberespionage, which not only compromises national security but also jeopardizes privacy on a global scale.

Sabotage, another key element of cyberwarfare, entails targeting and incapacitating critical infrastructure such as power grids or financial networks. A notable example is the Stuxnet attack on Iran's nuclear facility in 2010, where cyber tools were used to damage Iranian nuclear centrifuges without leaving physical traces. Sampanis et al. (2023) point out that such attacks demonstrate how cyberwarfare can achieve real-world consequences—such as disruption of vital infrastructure—without any conventional military action.

Propaganda, often an overlooked aspect of cyberwarfare, refers to the deliberate spread of disinformation to influence public opinion, disrupt social stability, or destabilize political systems. As Bardin (2025) notes, these tactics can manipulate information, mislead the public, and create discord, as seen in Russian interference in U.S. elections via social media campaigns.

2.1.4 Technological Methods In Cyberwarfare

Cyberwarfare encompasses a wide array of tools, many of which originate from or resemble criminal hacking techniques but are often adapted with greater sophistication for state objectives. Key methods include:

- I. **Malware:** A broad category including worms, viruses, trojans, and spyware. Stuxnet, Flame, and Snake are notable examples designed for sabotage or espionage. Some, like Shamoon or DarkHotel, act as “wipers,” erasing data to disable systems.
- II. **Ransomware & Data Wipers:** These attacks encrypt or destroy data, either for extortion or strategic disruption. While tools like WannaCry or NotPetya appear criminal, they've often been traced to state actors or used for geopolitical purposes. Colonial Pipeline's 2021 shutdown by ransomware revealed how digital extortion can have international ripple effects.

- III. **Distributed Denial-of-Service (DDoS):** Overloading systems with traffic to knock them offline, often used in conjunction with other attacks. DDoS campaigns have been deployed from Georgia (2008) to Ukraine (2022), targeting everything from courts to media outlets.
- IV. **Zero-Day Exploits:** Vulnerabilities unknown to the public and sometimes even to software developers are prized by state hackers for their stealth. Tools like EternalBlue (later exploited by WannaCry) show how these flaws can be weaponized globally.
- V. **Phishing & Social Engineering:** Human error remains a common entry point. Spear-phishing campaigns like those behind the SolarWinds breach (2020) and DNC hacks (2016) demonstrate how crafted emails can compromise even high-security systems.
- VI. **Influence Operations & Disinformation:** Troll farms and botnets are used to spread fake news, erode trust, and inflame divisions. The Internet Research Agency's campaigns in the U.S. and Europe exemplify how digital information manipulation can be strategically deployed.

Other Methods: Advanced cyberwarfare may involve tampering with supply chains, satellite networks, and critical infrastructure. Attacks on Viasat modems during the Ukraine war and emerging AI-driven malware illustrate how the battlefield continues to evolve.

2.1.5 PRIVACY AND SURVEILLANCE IN THE DIGITAL AGE

Perhaps the most persistent consequence of cyberwarfare is the erosion of privacy. As cyber operations become normalized, civilian data is increasingly swept up in surveillance programs or exposed through espionage. Governments cite national security, but the result is often broad, unchecked data collection.

The Snowden revelations (2013) revealed extensive global surveillance programs, prompting debate over the legality and ethics of mass data collection. Agencies like the NSA and GCHQ collected phone records, emails, and other communications—often without public knowledge or consent.

Meanwhile, cyberspace has become a battlefield, where individuals become collateral victims of digital espionage. Facial recognition, location tracking, and algorithmic profiling have become common, especially in regimes prioritizing control over civil liberties.

As cyberwarfare intensifies, privacy risks will only grow. Democratic societies must navigate the delicate balance between defending against digital threats and protecting civil liberties. Without updated legal frameworks and strong oversight, the militarization of cyberspace threatens the very freedoms it aims to defend.

2.1.6 Privacy and Surveillance

Privacy in the digital era encompasses an individual's right to govern their personal data, how it's collected, stored, and disseminated. It involves safeguarding against unauthorized surveillance, data leaks, and invasive monitoring practices. As highlighted by Dogbey (2023), protecting privacy is critical to upholding personal autonomy, dignity, and security.

Global human rights instruments affirm privacy as a core right. Key documents like the Universal Declaration of Human Rights (Article 12) and the International Covenant on Civil and Political Rights (Article 17) stress the importance of shielding individuals from arbitrary or unlawful intrusions into their private lives.

Nevertheless, a persistent tension exists between the imperatives of national security and the preservation of privacy. Efforts to bolster cybersecurity frequently involve comprehensive surveillance systems intended to detect threats, yet these measures can deeply encroach on individual privacy. The disclosures by whistleblower Edward Snowden revealed the vast reach of surveillance operations by agencies such as the NSA, igniting widespread debate over how to ethically balance state security and personal freedoms.

2.1.7 Privacy And Civil Liberties Concerns

The tension between cybersecurity and privacy is a growing concern. As states ramp up efforts to protect themselves from cyber threats, they often justify heightened

surveillance, data collection, and monitoring of digital communications. However, this can infringe on personal privacy and civil liberties. Libicki (2016) warns that while robust cybersecurity measures are necessary, they often involve the monitoring of network traffic and the collection of data, which can violate individuals' privacy.

The Snowden revelations in 2013 exposed the scale of government surveillance programs, bringing attention to how personal data is collected under the pretext of national security. As Gervais (2022) argues, these surveillance programs often target not only potential threats but also innocent civilians, raising ethical questions about the balance between security and privacy. Cyberwarfare exacerbates this issue, as states may leverage wartime powers to expand surveillance capabilities, bypassing privacy safeguards and legal constraints.

Moreover, cyber-attacks themselves may deliberately target civilian data, often to disrupt or coerce populations. The Russian cyber-attacks on Ukraine during the 2014 conflict, for example, focused not just on military systems but also on civilian networks, affecting millions of people. This deliberate targeting of civilian data in warfare represents a disturbing new dimension of conflict, where individual rights are sacrificed in the name of national security.

The integration of cyberwarfare into modern conflicts raises critical concerns about privacy and civil liberties. As states increasingly rely on cyber tools to achieve strategic objectives, the risk of indiscriminate damage to civilian infrastructure and the erosion of privacy grows. The need for international frameworks to address these challenges is urgent, ensuring that digital rights and individual freedoms are safeguarded in the face of evolving threats.

2.1.8 AI IN CYBER ATTACKS

The convergence of Artificial Intelligence (AI) and cyberwarfare marks a significant transformation in both offensive and defensive operations. AI-powered cyberattacks are capable of learning, adapting, and autonomously responding to changing environments, making them more effective and harder to detect.

AI-driven tools can now identify vulnerabilities in real-time, generate zero-day exploits, and craft spear-phishing attacks tailored to specific individuals. This level of automation introduces a new level of sophistication, creating self-learning cyberweapons that evolve during the attack process.

At the same time, AI plays a crucial role in defense. AI-enhanced threat detection systems can identify anomalies, anticipate intrusions, and respond faster than human operators. This arms race between AI-powered offense and AI-powered defense is shaping the next frontier of cyber conflict.

As Robert J. Harkins (2022) notes, this dynamic complicates cyber defence planning, as attackers leverage AI to mask their operations and generate false positives, overwhelming human analysts. The balance between AI's utility and its unpredictability will be central to future national cybersecurity strategies.

2.1.9 Ai And Privacy Violations

Despite its benefits, AI also presents profound privacy challenges. Its ability to analyse vast data sets enables more personalized and invasive cyberattacks, increasing the risk of mass data theft. AI-powered tools can profile individuals, predict behaviours, and exploit digital footprints to penetrate private systems.

Governments are also turning to AI for surveillance, with tools capable of tracking communications, social media activity, and physical movements. In countries like China, facial recognition and predictive analytics are already in widespread use, raising serious concerns about digital authoritarianism.

As Annika Jenson (2023) explores, AI amplifies both surveillance capabilities and the erosion of civil liberties. The balance between national security and personal privacy is becoming increasingly tenuous, forcing a global reckoning on the ethical use of AI in the digital age.

2.1.10 Cyber Sovereignty And State Responsibility

Cyber sovereignty, also known as digital sovereignty, refers to a nation's authority to oversee and regulate cyberspace within its own borders. This principle includes the power to legislate digital behavior, defend national cyberspace from external threats, and control the flow of content and digital platforms. As Sampanis et al. (2023) observe, growing digital risks have led states to increasingly assert control over their cyberspace to safeguard national interests and ensure security.

States face a dual responsibility in the realm of cyber governance. On one hand, they are entitled to protect their digital infrastructure from foreign threats and cyberattacks. On the other, they are bound by international law to uphold the sovereignty and privacy of other nations and individuals. Balancing these obligations demands that states implement cybersecurity strategies that not only defend national interests but also respect human rights, uphold transparency, and remain accountable in their actions.

These intricate challenges have amplified global debates on the need for standardized, enforceable frameworks to guide responsible state behavior in cyberspace. Grasping these dynamics is crucial for analyzing how cyberwarfare impacts both international security and individual privacy on a global scale.

2.2 THEORETICAL FRAMEWORK

2.2.1 Realism And Cyberwarfare

Realism, a foundational theory in International Relations, asserts that states operate in an anarchic international system where the primary pursuit is power and security. Thinkers such as Hans Morgenthau and Kenneth Waltz argue that states are driven by the imperative to safeguard their survival and national interests by enhancing their relative power. Within this view, international politics revolves around competition, power rivalries, and security dilemmas where efforts by one state to improve its own security can inadvertently threaten others, thereby heightening tensions and sparking rivalry (Waltz, 1979).

When applied to cyberwarfare, realism offers valuable insights into state behaviour in the digital realm. Cyber capabilities serve as strategic tools that allow states to project influence and defend interests without resorting to conventional military force. Even less powerful states can leverage digital tools to challenge dominant powers, thereby reshaping traditional power hierarchies. These dynamics exemplify realism's security dilemma: one state's cyber defence measures may be interpreted by others as hostile, prompting retaliatory cyber buildups. The global race for cyber dominance mirrors realism's portrayal of a relentless struggle for power and security in international affairs (Nye, 2022).

2.2.2 **Securitization Theory**

Securitization theory, developed by Barry Buzan, Ole Wæver, and Jaap de Wilde (1998), explores how states frame certain issues as security threats requiring urgent and exceptional action. A topic is considered "securitized" when state actors convincingly portray it as an existential threat, thereby legitimizing extraordinary policy measures. This often includes persuading the public and policymakers that unusual actions such as expanded surveillance powers or reduced transparency are warranted to defend national security.

In the sphere of cyberwarfare and digital surveillance, securitization theory helps explain how governments rationalize broad cybersecurity initiatives. Cyber threats are often portrayed as existential dangers, enabling states to justify sweeping data collection and aggressive monitoring practices. As Balzacq (2011) notes, these narratives facilitate public support for enhanced governmental authority in cyberspace. The Snowden disclosures underscored how national security rhetoric can successfully securitize digital threats, legitimizing far-reaching surveillance programs even amid public unease over civil liberties (Greenwald, 2014).

2.2.3 **Just War Theory (Jus Ad Bellum, Jus In Bello)**

Just War Theory offers a moral framework for evaluating the justification and conduct of war, distinguishing between when it is right to go to war (*jus ad bellum*) and how war should be conducted (*jus in bello*). Central principles include legitimate authority,

just cause, proportionality, and discrimination between combatants and non-combatants (Orend, 2013).

In the realm of cyberwarfare, this theory raises pressing ethical questions. Jus ad bellum criteria demand that cyberattacks be state-authorized and undertaken for morally justifiable reasons, such as self-defence. Responses must also be proportionate to the threats posed (Dipert, 2010).

Jus in bello principles guide ethical behaviour during cyber operations, requiring proportionality in the use of force and clear differentiation between military and civilian targets. Cyberattacks must avoid unnecessary harm, particularly to civilian infrastructure. Yet, as Taddeo (2017) highlights, cyber operations often blur these distinctions, making it difficult to apply conventional just war standards. These dilemmas lie at the heart of ethical debates surrounding cyber conflicts, where the digital nature of attacks complicates moral accountability.

2.2.4 Legal And Ethical Challenges

Cyberwarfare exists in a murky legal landscape. The UN Charter prohibits the use of force (Article 2(4)), and Article 51 allows self-defence but it's unclear whether a cyberattack without physical destruction qualifies as an "armed attack." The Tallinn Manual offers guidance: if a cyber operation disables military capability, it might count. But espionage, even via hacking, is not considered a use of force and remains largely tolerated in international law.

Sovereignty violations, however, are more clearly prohibited. States are not allowed to operate on another's digital infrastructure without consent. Yet, attribution remains a problem many attacks go unclaimed, allowing states plausible deniability.

International humanitarian law (IHL) also applies during war. According to the ICRC, cyber weapons must adhere to principles of distinction, proportionality, and military necessity. Civilian systems like hospitals or power grids should be off-limits, but dual-use networks complicate these standards in practice.

Outside of conflict, human rights law remains relevant. The ICCPR (Article 17) protects privacy, and the UNHRC affirms online freedoms. But global surveillance, as revealed by Snowden, often skirts these norms. Jurisdictional hurdles further complicate accountability hackers operate across borders, laws like the CLOUD Act or GDPR offer partial remedies, but enforcement is fragmented.

Meanwhile, national doctrines diverge. The U.S. Cyber Strategy (2023) supports “defend forward” tactics pre-emptively disrupting threats abroad. China’s cyber sovereignty model emphasizes state control, while Russia’s hybrid war doctrine blends information and cyber tools in asymmetric conflict. These competing views shape how laws are applied or ignored.

Ethically, the rise of patriotic hackers, cyber mercenaries, and AI-driven operations raises unresolved questions about combatant status and accountability. If civilians launch attacks, do they become lawful targets? What protections apply to contractors who develop military-grade malware?

In sum, international law remains behind the curve. While norms exist, binding rules and enforcement mechanisms are limited. The digital battlefield is expanding faster than legal consensus can catch up.

Understanding these theoretical frameworks realism, securitization theory, and Just War Theory is essential for analysing the complex landscape of cyberwarfare. Each provides a unique lens through which to interpret state actions, ethical challenges, and the shifting dynamics of international security shaped by digital conflict.

2.2.5 Cybercrime As A Tool For Cyberwarfare

Cyberwarfare has become a multifaceted arena where the boundaries between statecraft and cybercrime are increasingly blurred. In recent years, nation-state actors have strategically adopted cybercriminal tactics to advance geopolitical objectives, leveraging cyber-attacks for espionage, financial disruption, and the destabilization of rival economies all without direct military engagement.

A key example is the Bybit hack (2025), attributed to the Lazarus Group, a North Korean-backed cybercriminal organization. The theft of approximately \$1.5 billion in cryptocurrency highlights how cybercrime is now integral to state-sponsored operations, serving both as a source of funding and as a tool for economic destabilization. The Lazarus Group exemplifies the rise of proxy cybercriminals, operating under state guidance to erode public trust in financial systems and weaken national resilience.

As Michael G. Kearns (2020) argues, the use of cybercrime in statecraft extends well beyond theft it's a strategic tool used to manipulate financial systems, steal intellectual property, and exert political influence. As states increasingly weaponize digital criminal networks, cybercrime has become an indispensable element of modern conflict, presenting new challenges for regulation and attribution.

2.2.6 Ransomware And Its Geopolitical Implications

Once considered a tool for pure financial gain, ransomware has evolved into a powerful instrument of cyberwarfare, targeting not just private enterprises but also critical national infrastructure.

The WannaCry attack (2017), linked to North Korea, and the NotPetya incident (2017), attributed to Russia, illustrate this shift. While WannaCry affected healthcare and educational systems worldwide, NotPetya was a politically motivated attack aimed at crippling Ukraine's economy, causing billions in global damages.

More recently, the Colonial Pipeline attack (2021) demonstrated how ransomware can paralyze essential infrastructure, induce public panic, and cause economic disruptions—effects rarely achieved through traditional military means. These cases reflect how ransomware, when used by state actors, transcends criminality to become a geopolitical weapon.

With increasing sophistication, ransomware presents a dual-use dilemma: it serves both as a tool of financial extortion and a mechanism for state-sponsored sabotage. Its rise demands urgent attention to the legal and regulatory frameworks needed to

respond to cyber aggression in an age where code can cause chaos without firing a single shot.

2.2.7 Hactivism And Cyberterrorism

The rise of non-state actors in cyberwarfare introduces additional complexity. Groups like Anonymous and LulzSec use hacking as a form of political or social protest, known as hactivism. Though often non-lethal, these actions can disrupt institutions, expose sensitive information, and influence public discourse.

In contrast, cyberterrorist groups aim to cause fear, economic damage, or societal instability sometimes aligning with physical acts of terror. Their decentralized nature and ideological motives make attribution and deterrence particularly difficult.

As David H. Daugherty (2021) notes, these actors challenge traditional definitions of warfare and raise questions about legal classification. Should hactivists be treated as combatants, criminals, or something in between? These ambiguities complicate global response mechanisms and laws of armed conflict.

2.2.8 Cyberwarfare And Proxy Conflicts

Increasingly, states are using cyber operations in proxy conflicts, employing third-party actors to avoid direct confrontation. These indirect engagements allow for plausible deniability while achieving strategic aims.

Russia's use of cyberattacks during the 2014 annexation of Crimea and throughout the Ukraine conflict exemplifies this approach. States can also hire cyber mercenaries, conduct operations via private contractors, or rely on semi-autonomous hacker groups to carry out politically motivated missions.

Laura B. Thompson (2022) emphasizes that cyber proxies now play a central role in global conflict, offering states a low-cost, high-impact alternative to traditional warfare. However, their involvement muddies legal accountability and increases the risk of escalation in digital conflicts.

2.3 EMPIRICAL FRAMEWORK

2.3.1 REVIEW OF RELATED WORKS

A thorough understanding of related studies is crucial for research, as it forms the foundation for future exploration. This understanding enables researchers to grasp what has already been achieved in the field and to pinpoint the knowledge gaps that their current research seeks to fill. By building on prior work, researchers can ensure that their studies are relevant, well-informed, and contribute significantly to the existing body of knowledge. This section presents background studies and explores further the reviews that are directly related to the objective of this research.

2.3.2 Offensive Cyber Capabilities and State Violence (Freire, 2023)

Freire's examination of offensive cyber capabilities highlights a significant shift in the nature of modern warfare. Traditionally, warfare was confined to physical domains i.e. land, sea, and air where states employed military forces to secure their interests. With the rise of the digital age, however, cyber tools have become integral to state power, enabling states to influence, undermine adversaries, and conduct covert operations. Freire's paper underscores how cyberwarfare has evolved from a peripheral tactic to a mainstream tool of state violence.

Freire challenges the conventional view of cyberwarfare as an asymmetric tactic used primarily by weaker states or non-state actors. Rather, he argues that cyberwarfare has been institutionalized as a core element of state strategy, particularly among powerful nations. Cyber capabilities are now used not just for military purposes but also to disrupt economic systems, infiltrate critical infrastructure, and even manipulate social structures. This normalization of cyber tools in statecraft significantly shifts the boundary between wartime and peacetime, making it increasingly difficult to determine when a conflict truly begins or ends. In this sense, cyberwarfare has become a strategic lever for altering geopolitical power.

One of the most profound consequences of this shift is the erosion of privacy. Cyberwarfare often targets not just military or governmental infrastructure but also the personal data of civilians. Increasingly, state-sponsored cyberattacks involve mass

surveillance, data breaches, and digital espionage, all of which lead to privacy violations on a massive scale. States often justify these breaches under the banner of national security, a rationale that, as Freire observes, frequently shields governments from international accountability. The theft of personal data, the hacking of private communications, or the manipulation of social media platforms are examples of how cyberwarfare directly impacts individuals' privacy rights.

Freire also brings to light the ethical concerns surrounding cyberwarfare. As cyber capabilities advance, they increasingly blur the line between defensive actions and aggressive warfare. States often present cyberattacks as measures to protect national security. However, the widespread erosion of privacy raises important ethical questions, especially regarding the unintended harm caused to non-combatants and the broader civilian population. Freire notes that the distinction between legitimate defence and covert aggression is becoming increasingly difficult to discern, particularly when cyber tools are used in non-attributable, covert operations that are hard to trace.

Freire's work offers a critical perspective on the ambiguity surrounding cyberwarfare. By targeting civilian data and critical infrastructure, states engage in practices that not only violate privacy but also undermine public trust in digital systems. This blurring of lines between security and aggression presents serious challenges for both international security and the protection of individual privacy. Freire's analysis of the lack of accountability in cyberwarfare highlights how states can wield cyber capabilities with little consequence, often using national interest as a justification.

The paper's insights are crucial to understanding the intersection of cyberwarfare and privacy deprivation. As states continue to justify their actions as necessary for national defense, the erosion of privacy is becoming a pervasive issue, often presented as a cost of securing the nation. This ethical dilemma is at the heart of this study, urging policymakers and legal experts to confront the question: How can states protect cybersecurity without compromising individual privacy in the digital age? Freire's exploration of the strategic use of cyber capabilities and its ethical ramifications serves as an essential reference for understanding the broader consequences of cyberwarfare on both state sovereignty and individual freedoms.

Freire also emphasizes that existing legal frameworks are ill-equipped to handle the complexities of cyberwarfare. While traditional laws of war, such as the Geneva Conventions, were designed for physical conflicts, they fail to address the challenges posed by digital warfare. Cyberwarfare operates in a legal gray area, where actions that would be considered acts of war in traditional settings may not be recognized as such in cyberspace. This lack of clarity allows states to engage in cyberwarfare without facing the same legal consequences as they would for conventional aggression. Freire argues that this legal ambiguity is a significant challenge, allowing states to use cyber capabilities for both offensive and defensive purposes with minimal legal oversight.

Freire's work offers a compelling argument for why cyberwarfare represents a dual challenge: it provides states with new ways to defend their sovereignty and project power, but it also threatens individual privacy and human rights on a global scale. As the digital domain becomes a critical space for state security, it is essential to reassess the balance between national defense and the protection of personal privacy. Freire's research underscores the urgent need for international cooperation and the creation of legal frameworks that regulate cyberwarfare and safeguard privacy rights in an increasingly digitized world.

Freire's paper sheds light on how the strategic use of offensive cyber capabilities has transformed state violence, introducing complex ethical and legal challenges for both national security and individual privacy. As states increasingly adopt cyberwarfare as a cornerstone of their military strategies, the critical issue of how to protect personal privacy while addressing security needs will remain a key concern in international relations and digital governance. The analysis calls for updated international legal standards to effectively address the complexities of cyber conflicts and to ensure privacy rights are maintained in the digital age.

2.3.3 Human Security And Peace In Cyberspace (Dogbey, 2023)

In his paper, Dogbey addresses a critical aspect of cyberwarfare its impact on human security which extends beyond state infrastructure to directly threaten the safety and privacy of individuals. As digital warfare and surveillance techniques advance, cyber tools provide states with the capability to monitor, intercept, and manipulate private

data, often without the knowledge or consent of those affected. Dogbey argues that cyberwarfare is no longer a threat solely to national security; it is increasingly a human security issue as well, directly affecting the rights and freedoms of individuals.

What stands out in Dogbey's analysis is the relationship between cyber-attacks and the erosion of privacy. While states justify cyber operations as vital for national defense, Dogbey highlights the ethical dilemma this creates: how can governments protect national security without violating the privacy rights of their citizens? He suggests that cyberwarfare not only harms those directly targeted by cyber-attacks but also creates a broader societal impact. The "chilling effect" a term Dogbey uses to describe how individuals hesitate to freely use digital platforms due to fear of surveillance undermines public confidence in online systems, weakening democratic engagement and civil liberties.

Dogbey's work provides a solid framework for understanding the human cost of cyberwarfare, particularly the infringement on privacy rights in the name of security. This perspective is highly relevant to my research, as it highlights the tension between national security goals and individual freedoms, a central issue in the broader conversation about cyberwarfare. Dogbey calls for international collaboration and the establishment of global norms to ensure that while states protect their interests, they do not infringe upon fundamental human rights. This call for cooperation resonates deeply with my study, emphasizing that the challenges of cyberwarfare are global in scope and must be addressed through collective action, rather than isolated national responses.

2.3.4 Cyberwarfare In The Modern World (Sampanis Et Al., 2023)

In their 2023 paper, Sampanis, Spiliopoulou, and Maglaras present a nuanced and comprehensive analysis of cyberwarfare as it is used by states to secure strategic interests and undermine their adversaries. The authors argue that while cyberwarfare has become an integral part of modern conflict, it remains an evolving and multifaceted challenge, especially when it comes to the legal and ethical dimensions. This paper is particularly significant for understanding the dual nature of cyberwarfare—on one hand, it is an effective strategic tool for states to exert influence and project power,

and on the other hand, it raises serious concerns about privacy violations and the erosion of digital rights.

They began by addressing how cyberwarfare has fundamentally changed the landscape of conflict in the 21st century. Unlike traditional warfare, which is usually fought in physical spaces such as land, air, and sea, cyberwarfare unfolds in the digital domain, making it a less predictable and more covert form of conflict. The authors explain that cyber-attacks can target critical infrastructure, economic systems, and even societal values, which makes them attractive to states as they provide a relatively low-cost, high-impact means of influencing or destabilizing adversaries. However, as states increasingly adopt cyberwarfare as part of their defence and offense strategies, the consequences for privacy are often overlooked.

What is particularly compelling in Sampanis et al.'s paper is the observation that cyber-attacks do not respect national borders, and thus, attribution of these attacks becomes a major challenge. Cyber operations often involve anonymous actors, whether state-sponsored or non-state entities, making it difficult to definitively identify the responsible party. This lack of accountability creates a legal vacuum in which states can carry out cyber-operations that violate international norms with minimal fear of repercussion. This situation is problematic, particularly in the context of privacy rights, as cyber-attacks increasingly target not just military or political data, but also personal information from private citizens. Sensitive data such as financial records, medical histories, and private communications are often caught in the crossfire of these digital conflicts, leading to privacy breaches that affect millions of innocent people globally.

The authors draw attention to this discrepancy between state security needs and personal privacy—while states argue that such attacks are necessary for maintaining national security, the erosion of privacy can lead to widespread harm. The targeting of private data in the context of cyberwarfare becomes a significant ethical dilemma, as it raises critical questions about the balance between state interests and the fundamental rights of individuals. Sampanis et al. push for more global regulatory frameworks to address these concerns, advocating for policies that protect privacy

while ensuring that states have the tools they need to defend themselves in the digital era.

Another key insight from this paper is the authors' discussion of the power asymmetry inherent in cyberwarfare. Traditional military conflicts often favour states with larger, more advanced military capabilities. However, cyberwarfare levels the playing field, allowing smaller states or even non-state actors to carry out disruptive cyber-attacks that can have significant impact on much larger, more powerful nations. This asymmetric nature of cyberwarfare makes it a more accessible form of conflict for states with fewer resources or even for hacktivist groups and terrorist organizations. For these actors, cyberwarfare provides a way to challenge established powers without the need for conventional weapons or large-scale infrastructure.

While this democratization of conflict might seem like a positive development in terms of reducing global inequalities, it also has serious implications for privacy and digital rights. As smaller states or non-state actors engage in cyber-attacks, the risk of unintended consequences increases—civilian populations in adversary nations may be exposed to massive breaches of privacy due to poorly targeted cyber operations or the exploitation of personal data for political purposes. This power imbalance, while enabling less powerful actors to compete in a new digital theater, has profound implications for the protection of privacy. What is particularly troubling is the increasing incursion of private data into the realm of geopolitical conflicts, where individuals may suffer without any direct involvement or understanding of the underlying conflict.

The paper also addresses the lack of legal frameworks that specifically govern cyberwarfare. Traditional laws of war, such as the Geneva Conventions, were crafted with physical warfare in mind and do not adequately address the complexities of cyber operations. This lack of relevant legal guidance means that states can engage in cyberwarfare with little regard for privacy rights or human rights, particularly when the attacks are carried out under the guise of national defence.

Sampanis et al. call for the creation of more robust international frameworks to regulate cyber conflicts and protect individuals' privacy in the face of rising cyber

threats. They argue that it is imperative for the international community to come together and develop cyber laws that are specifically tailored to the digital age, offering guidelines for when and how states can engage in cyber-operations. These frameworks would need to address the protection of civilians in times of cyber conflict, ensuring that the privacy of individuals is not unduly compromised during state-led or state-sponsored cyber operations.

The authors stress the importance of defining clear boundaries in the digital realm, where military objectives must be separated from the protection of civilian life and personal data. This, they argue, is crucial for preventing cyberattacks from becoming a free-for-all that undermines both national security and the privacy rights of individuals.

What truly stands out in this paper is how Sampanis et al. frame cyberwarfare as a challenge that is not just a matter of security or military strategy, but one that directly impacts individual privacy and digital rights. Cyberwarfare may seem like a distant issue, confined to the realm of state actors and large-scale geopolitical conflicts, but in reality, its consequences affect ordinary citizens who have no involvement in these conflicts. The authors emphasize that cyberwarfare raises critical ethical and personal considerations, particularly in regard to the right to privacy, which is becoming increasingly fragile in the digital world.

They argue that as states continue to develop their cyber capabilities, they must be mindful of how these capabilities are used, and ensure that they do not infringe upon the personal freedoms of citizens, either within their borders or beyond them. The authors suggest that cyberwarfare cannot simply be seen through the lens of state interests but must be understood as part of a broader conversation about the digital rights of individuals in the 21st century. This perspective adds depth to the study, reinforcing the need for balanced, ethical considerations in both the development and deployment of cybersecurity measures.

The paper offers a critical examination of cyberwarfare as it intersects with privacy and digital rights. The authors highlight the ethical complexities and legal challenges of this

new form of warfare, especially when it involves privacy violations. The discussion of power asymmetry, where smaller states or non-state actors can disrupt larger nations, and the lack of accountability for cyber-attacks, particularly in regard to privacy violations, brings to light the urgent need for international regulation. Their call for robust frameworks to govern cyberwarfare and protect digital rights is a pivotal insight for this research, as it reinforces the argument that while states have legitimate concerns regarding cybersecurity, there must be a clear distinction between defending national interests and violating personal privacy.

In essence, the paper deepens the understanding of cyberwarfare as not just a security issue but as a complex human rights challenge, with broad implications for the protection of privacy in a digital world. It underscores the need for legal innovation and international collaboration to create frameworks that respect both state security and individual freedoms in the digital age.

2.3.5 Cyber Attacks And The Laws Of War (Gervais, 2022)

Gervais's analysis of cyberwarfare presents a critical examination of the growing inadequacy of traditional legal frameworks, particularly the Geneva Conventions, in addressing the complexities of digital conflict. As warfare shifts from physical confrontation to digital operations, international law has struggled to keep pace, especially when it comes to privacy violations and individual rights. Gervais argues for the urgent need to redefine how cyberwarfare is regulated, advocating for a system that balances national security concerns with the protection of civilian rights, especially privacy.

At the heart of Gervais's critique is the disconnect between conventional warfare laws, designed for physical spaces, and the virtual realm of cyberattacks. The Geneva Conventions were created to govern traditional warfare, aiming to protect civilians and minimize destruction during conflicts. However, they were framed in an era where combat took place in clearly defined spaces with tangible weapons. In contrast, cyberattacks are inherently ambiguous and often occur in virtual spaces, where targets can be unclear and the effects can reach across borders with global consequences.

Gervais highlights the unique challenges cyberwarfare presents to existing legal standards. Traditional concepts like military targets, combatant status, and civilian protection become muddled when states use cyber tools to disrupt systems on a global scale. A state might launch a cyber-attack aimed at military infrastructure, but the same operation could also breach the personal privacy of civilians around the world, an impact that traditional warfare laws do not address.

Privacy Violations in Cyberwarfare One of Gervais's most pressing concerns is the privacy violations resulting from cyberwarfare. Cyber tools enable states to access, manipulate, and steal personal data, from communications and financial records to healthcare information. These breaches often affect civilians far removed from the actual conflict. Unlike traditional warfare, where damage is primarily physical, cyberattacks often undermine privacy rights on an unprecedented scale. The intrusion into personal data—whether through hacking, surveillance, or espionage—can severely impact individuals' autonomy, financial stability, and reputations.

Gervais underscores that while states often justify these breaches as necessary for national security, the ethical and legal dilemmas posed by such actions are significant. Cyberattacks may not result in immediate physical destruction, but the long-term damage to citizens' trust in digital systems and their privacy is profound. Manipulated or stolen personal information can have lasting consequences for individuals' lives, especially when it is used for purposes such as surveillance, blackmail, or financial fraud.

The Need for New International Legal Frameworks Gervais stresses that current international legal frameworks are inadequate to regulate cyberwarfare, particularly in the area of privacy protection. Traditional laws like the Geneva Conventions were designed for physical warfare, and they do not adequately address the challenges posed by digital warfare. Cyberwarfare presents new obstacles, such as the difficulty of attribution, the asymmetric nature of attacks, and the invisible impact on civilians that current laws fail to cover. The lack of clear legal definitions and standards around cyber-attacks leaves a legal grey area, making it difficult to apply traditional rules of engagement to digital warfare.

To address this, Gervais advocates for the creation of new international laws specifically designed to address the realities of cyberwarfare. These laws should establish clear definitions of what constitutes a cyber-attack in the context of armed conflict, as well as guidelines for when and how cyber tools can be deployed. Importantly, these frameworks should also prioritize the protection of privacy and regulate the ethical use of cyber capabilities during conflicts. Gervais calls for international treaties or agreements that would bind states to adhere to specific norms governing the use of cyber tools in times of conflict, ensuring that such actions do not result in excessive violations of privacy.

Cyberwarfare as an Ethical and Human Rights Issue In addition to legal concerns, Gervais emphasizes that cyberwarfare is also fundamentally an ethical and human rights issue. Traditional warfare laws were founded on the humanitarian principle of minimizing civilian harm and protecting the dignity of individuals. However, cyber-attacks can have far-reaching consequences for individual freedoms and privacy. Without adequate safeguards, cyberwarfare threatens not only national security but also fundamental human rights, such as the right to privacy, freedom of expression, and autonomy.

Gervais points out that cyberwarfare undermines public trust in digital systems, particularly when states engage in surveillance or manipulate digital platforms. The global nature of cyberwarfare complicates these ethical issues, as an attack launched in one country can affect individuals in other countries, violating privacy rights and impacting civil liberties across borders. Gervais argues that in the digital age, the responsibility to protect individuals' rights transcends national boundaries, and international laws must evolve to address this new reality.

Recommendations for Policy and Legal Reforms Gervais proposes several reforms to address these concerns. First, he advocates for clear definitions of what constitutes a cyber-attack in the context of war and rules of engagement that prioritize privacy protection and safeguard civilians from harm. Additionally, he suggests the creation of accountability mechanisms, such as third-party oversight or the establishment of international cyber tribunals, to assess the legality and ethics of cyber operations that

impact civilian infrastructure or result in privacy breaches. Gervais's work offers crucial insights into the legal and ethical challenges posed by cyberwarfare, particularly the privacy violations that often accompany cyber-attacks. His call for the development of new international frameworks is essential for ensuring that cyberwarfare is governed by clear and enforceable laws, particularly when it comes to protecting individual privacy and human rights in the digital age.

The literature reviewed consistently reinforces the idea that cyberwarfare is not just a tool of military strategy, but also a growing human rights concern. For example, Freire emphasizes how states are increasingly leveraging cyber capabilities to assert their power, often using digital attacks that target civilians. Similarly, Dogbey highlights how cyberwarfare poses significant risks to human security, underlining that it's not just about geopolitical conflict, it's about safeguarding the rights and freedoms of everyday individuals. Across these studies, a recurring theme emerges: cyberwarfare creates ethical dilemmas and leads to privacy violations that cannot be ignored.

Each of these papers brings a distinct perspective on how states deploy cyber tools, yet they all share a common focus individual privacy. There's growing consensus in the literature that current legal frameworks are insufficient to address the complex nature of cyber threats. This gap highlights the urgent need for stronger, more comprehensive regulations to protect civilians from the unintended consequences of state-backed cyber actions.

What stands out most in these discussions is the call for global cooperation. As cyber threats grow more frequent and sophisticated, the need for international regulations becomes clearer than ever. The challenge lies in ensuring that cybersecurity efforts are balanced with the protection of civil liberties and privacy rights. It is increasingly evident that moving forward, there must be a framework that guarantees national security efforts do not infringe upon personal freedoms, urging the creation of rules that govern not just military action but also the protection of human rights in cyberspace.

CHAPTER 3 RESEARCH METHODOLOGY

This chapter outlines the methodology employed in this research. It covers the research approach, the case study selection process, data collection and analysis techniques, and the rationale behind choosing a qualitative research design. Given the complexity of the topic, this study uses thematic analysis of documented case studies to explore the different tactics used in cyberwarfare and their impact on privacy and civil liberties. This chapter provides the foundation for understanding how data were gathered, analysed, and interpreted to identify broader trends in cyberwarfare and privacy deprivation.

3.1 RESEARCH APPROACH

This research adopts a qualitative, exploratory approach tailored to the complexities of cyberwarfare, which often involves classified information and rapidly evolving threats. Qualitative methods are suitable for capturing the nuanced socio-technical and legal aspects of cyberwarfare, especially where statistical data may be incomplete or unavailable. The study relies on thematic analysis to identify recurring patterns, motivations, and outcomes in documented cyberwarfare events from 2015 to 2025. The approach enables in-depth exploration of both strategic impacts (on national security and policy) and personal consequences (privacy breaches, surveillance).

3.2 CASE STUDY DESIGN

A multiple-case study design is employed to provide a holistic view and enable comparison across different incidents. The cases are selected based on their scale, level of documentation, and their relevance to both national security and privacy. This design facilitates the identification of common tactics, escalation paths, and outcomes that may inform future policy or technical responses.

Case Study Selection Criteria:

- I. Involvement of at least one nation-state actor or government-backed entity.

- II. Clear evidence of impact on either national infrastructure, governmental policy, or civilian privacy.
- III. Availability of multi-source documentation (academic, government, private sector reports).
- IV. Relevance to the 2015–2025 timeframe and global significance.

Selected Case Studies:

- 1. Ukraine-Russia Cyberwar (2015–2023)
- 2. SolarWinds Hack (2020)
- 3. Colonial Pipeline Ransomware Attack (2021)
- 4. U.S. 2016 & 2020 Election Interference
- 5. Bybit \$1.5B Hack (2025)
- 6. Ethiopia–Egypt Cyber Attacks On The Grand Ethiopian Renaissance Dam (Gerd) – 2020

The diversity of these cases allows for comparative analysis of different methods (malware, ransomware, supply-chain attack, disinformation), varying motivations (disruption, espionage, economic sabotage), and a spectrum of privacy outcomes.

3.3 DATA SOURCES

This study draws on a wide range of reputable secondary sources to ensure objectivity and credibility:

- I. **Governmental Reports:** U.S. Cybersecurity and Infrastructure Security Agency (CISA), European Union Agency for Cybersecurity (ENISA), National Cyber Security Centre (UK), NATO briefings, etc.

- II. **Academic Journals:** Peer-reviewed articles on cyberwarfare, privacy, and digital policy, published between 2015 and 2025.
- III. **Investigative Journalism and Media:** International outlets with strong technology and security reporting (e.g., Reuters, The Guardian, Wired, New York Times).
- IV. **Cybersecurity Industry Reports:** Threat intelligence and forensic analysis from leading firms (e.g., FireEye/Mandiant, CrowdStrike, Kaspersky, Dragos, Symantec).
- V. **Legal and Policy Publications:** Reports from international legal bodies, think tanks, and regulatory agencies on privacy, surveillance, and cybersecurity law.

All sources are cited in accordance with academic standards, and their reliability is cross-verified where possible.

3.4 DATA COLLECTION AND ANALYSIS

Data Collection:

Relevant documents, reports, and articles are collected through academic databases (Google Scholar, JSTOR, IEEE Xplore), government and agency portals, industry whitepapers, and major news archives. Selection prioritizes materials that provide clear, detailed accounts of cyber incidents, attack methodologies, and privacy consequences.

Thematic Coding and Analysis:

Collected data is systematically coded to identify recurring themes and categories:

- a. **Type of Attack:** Espionage, sabotage, ransomware, supply-chain compromise, disinformation.

- b. **Attack Vector/Methodology:** Malware, phishing, DDoS, zero-day exploitation, credential theft.
- c. **Impact Sector:** Government, critical infrastructure, elections, financial services, healthcare, etc.
- d. **Privacy Implications:** Scope and type of data breached, evidence of mass surveillance, policy responses, cross-border effects.

The analysis proceeds in stages:

1. **Descriptive Summaries:** Each case study is first summarized for factual accuracy and event chronology.
2. **Comparative Analysis:** Similarities and differences between cases are analyzed to draw broader insights into the tactics, impacts, and defensive responses observed across the decade.
3. **Triangulation:** Findings from different sources (academic, governmental, industry) are cross-referenced to reduce bias and improve validity.
4. **Synthesis:** Key patterns, lessons, and open questions are synthesized into the results and discussion chapters.

3.5 ETHICAL CONSIDERATIONS

This study relies solely on secondary and publicly available data, which means no direct contact with human subjects. Care is taken to avoid sensationalism, respect privacy, and use information ethically, particularly regarding personally identifiable information or classified intelligence. The research acknowledges the sensitivity of ongoing geopolitical conflicts and strives for impartiality, balancing technical analysis with social and legal context.

3.6 LIMITATIONS OF THE METHODOLOGY

While qualitative research and secondary data offer valuable insights, they also present limitations:

- a. **Source Reliability:** Some cyberwarfare incidents are underreported, politically sensitive, or subject to disinformation, which may affect the completeness or accuracy of available data.
- b. **Attribution Challenges:** Attribution of cyber operations to specific nation-states remains a debated and sometimes inconclusive process.
- c. **Evolving Threat Landscape:** The rapid evolution of cyber threats means that some tactics or impacts may not be fully captured by sources published before 2025.

These limitations are mitigated by cross-checking multiple independent sources, focusing on well-documented cases, and clearly stating any uncertainties in the discussion.

CHAPTER 4 IMPLEMENTATION/RESULTS AND DISCUSSION

This chapter presents a comprehensive analysis of the findings derived from the case study examination and thematic analysis of cyberwarfare's impact on national security and individual privacy. Drawing on selected incidents including the Stuxnet attack, the 2007 DDoS assault on Estonia, the Russia–Ukraine cyber conflict, and the Bybit \$1.5B hack attributed to the North Korean Lazarus Group, the analysis explores how cyberwarfare tactics are evolving, the strategic goals behind them, and their wider implications for both state defence systems and civil liberties.

The primary focus of this chapter is to explore the intersection of digital conflict, privacy violations, and the ethical, legal, and operational challenges arising from these increasingly complex threats. The chapter discusses the results of the thematic analysis in relation to the study's hypotheses, highlighting both the direct disruptions to critical infrastructure and the indirect consequences for individual freedoms in a rapidly digitizing security environment.

4.1 CYBERWARFARE AND NATIONAL SECURITY: STRATEGIC IMPLICATIONS

Once viewed as a peripheral element of modern conflict, cyberwarfare has now become a core component of statecraft and national security strategy. The case studies examined in this research reveal the increasing sophistication and intentional deployment of cyber operations by nation-states, underscoring their strategic value in shaping geopolitical outcomes.

4.1.1 UKRAINE–RUSSIA CYBERWAR (2015–2023)

The The Ukraine–Russia cyberwar is widely regarded as the most sustained and sophisticated campaign of state-backed digital aggression in modern history. What began as isolated intrusions in the mid-2010s has evolved into a relentless, multi-year

campaign of disruption, espionage, and hybrid warfare—intertwining military operations with cyber tactics that target every level of Ukrainian society.

The world's attention was first drawn to the severity of the Ukraine cyber conflict in December 2015, when a coordinated cyberattack led to a blackout for more than 200,000 people in Western Ukraine. The culprit: the BlackEnergy malware, wielded by the Russian-affiliated Sandworm group, which gained access to power utility control systems and remotely shut down substations (Dragos, 2017; Zetter, 2016). The attack was not just disruptive, it was unprecedented—the first time in history that hackers took down a power grid at scale. A follow-up strike in 2016 using the more advanced Industroyer malware underscored the adversary's growing technical prowess and willingness to target civilian infrastructure (Greenberg, 2019).

These incidents highlighted the vulnerabilities of critical infrastructure not just in Ukraine, but globally. The attacks were deeply psychological, designed to erode public trust in basic services, sow confusion, and demonstrate the reach of Russian cyber capabilities (ENISA, 2022). They also introduced a new sense of urgency among policymakers worldwide about the possibility and reality of “cyber-physical” warfare.

If BlackEnergy and Industroyer showed the potential for local devastation, the **NotPetya** incident in June 2017 illustrated how cyberwar can easily spiral into an international crisis. Originally seeded through a Ukrainian tax software update, NotPetya masqueraded as ransomware but was in fact a destructive wiper: its main purpose was to render computers inoperable, not collect ransom (Greenberg, 2018; BBC, 2017).

Within hours, NotPetya spread from Ukrainian banks, government agencies, and transportation networks to infect multinational corporations around the world. Global shipping giant Maersk lost access to its IT infrastructure, pharmaceutical leader Merck's manufacturing was halted, and countless other businesses suffered billions in losses. For Ukraine, the attack represented a further assault on its economy and state functionality; for the international community, it was a wake-up call that no digital

border is impermeable to the fallout of state-sponsored cyber conflict (Greenberg, 2018).

The full-scale Russian invasion of Ukraine in February 2022 marked a new chapter not just in conventional warfare, but also in the scale and coordination of digital operations. Leading up to and throughout the invasion, Ukraine faced waves of phishing campaigns, website defacements, Distributed Denial-of-Service (DDoS) attacks, and destructive malware (CISA, 2022; ENISA, 2022). Russian cyber actors, including Sandworm, attempted to paralyze governmental communication, erase or steal vital data, and sow panic among the civilian population. One of the most high-profile digital strikes was the attack on the Viasat satellite network, which disrupted both military and civilian connectivity during the invasion's first hours (ENISA, 2022; CISA, 2022).

However, Ukraine's response demonstrated impressive resilience. Drawing from years of experience, the country—supported by Western intelligence, private cybersecurity firms, and a “cyber volunteer army” of global hackers—was able to blunt the effectiveness of many attacks, restore critical systems, and even mount its own digital countermeasures (Greenberg, 2022). These efforts exemplified how “defensive alliances” can be forged quickly in the face of cyber aggression, and how civil society can play a direct role in modern conflict.

While the headlines often focus on grid failures and government websites, the privacy consequences of the Ukraine–Russia cyberwar have been equally profound. Cyberattacks have targeted government databases containing personal details, healthcare records, and financial information of millions of Ukrainians (ENISA, 2022). The psychological impact of knowing your communications, location data, or even hospital records might be exposed to hostile actors cannot be overstated. During wartime, fears of doxxing, blackmail, and targeted surveillance rose dramatically among activists, journalists, and ordinary citizens alike (Greenberg, 2022).

Additionally, Ukraine's defensive mobilization has sometimes required expanded government monitoring of digital activity, raising difficult questions about balancing

collective security and individual privacy rights (ENISA, 2022). Many privacy advocates worry that some surveillance powers, once justified by emergency, might persist even after the crisis passes.

The Ukraine–Russia cyberwar has forced a major reassessment of international norms. The frequency and scale of attacks violated the expectation that civilian infrastructure and bystanders would be protected in conflict (ENISA, 2022; CISA, 2022). Yet, attribution remained complex; states were often slow or cautious in formally blaming Russia for specific events due to diplomatic concerns or evidentiary uncertainty. This ambiguity, some argue, has emboldened further aggression.

The case has also catalyzed new policy debates in NATO, the EU, and global cybersecurity circles, spurring moves toward collective defense, real-time intelligence sharing, and the establishment of norms prohibiting attacks on healthcare, energy, and other critical sectors (ENISA, 2022).

In summary, the Ukraine–Russia cyberwar demonstrates how digital conflict has become deeply intertwined with modern warfare—impacting critical infrastructure, shaping international policy, and exposing millions to new privacy risks. It is a testament to the resilience and adaptability of societies under fire, but also a stark warning that the boundaries between war, crime, and civil life have never been less clear.

4.1.2 SOLARWINDS HACK (2020)

The SolarWinds hack, revealed to the world in late 2020, is widely regarded as one of the most consequential and sophisticated supply-chain cyberattacks in history. Unlike the more visible disruptions of power grids or ransomware shutdowns, the SolarWinds operation was marked by subtlety, patience, and the ability to compromise the very foundations of digital trust at the highest levels of government and industry (CISA, 2021; SANS, 2021).

The breach began when Russian state-sponsored threat actors attributed to the group known as APT29 or “Cozy Bear” managed to compromise the software development pipeline of SolarWinds, a Texas-based company whose Orion IT monitoring platform was used by thousands of organizations worldwide (CrowdStrike, 2021). Malicious code was covertly inserted into routine updates of Orion, and when customers installed these updates, the attackers gained a stealthy foothold inside their networks (CISA, 2021). What made the attack especially dangerous was the patience and technical sophistication: the backdoor, dubbed “Sunburst,” lay dormant for weeks or even months before being activated, allowing the attackers to carefully select their targets and exfiltrate sensitive data without raising suspicion (SANS, 2021).

SolarWinds’ customer list included U.S. federal agencies such as the Departments of Treasury, Commerce, Homeland Security, and Energy as well as technology giants like Microsoft and Cisco, critical infrastructure firms, and hundreds of Fortune 500 companies (CrowdStrike, 2021). At its peak, the breach is estimated to have given Russian intelligence access to over 18,000 networks (CISA, 2021). The true scope may never be fully known, since many victims were not directly named in public reports.

Importantly, the hack was not limited to data theft or simple espionage. The attackers gained the ability to monitor confidential communications, move laterally between systems, and even forge credentials that could be used for long-term surveillance or future attacks (CrowdStrike, 2021; SANS, 2021). The length of time the operation went undetected some estimates put it at nearly nine months highlighted how supply-chain attacks can be nearly invisible until discovered by accident or through an unrelated security investigation.

From a privacy perspective, SolarWinds was a nightmare scenario. Highly sensitive data government deliberations, corporate trade secrets, private emails was accessed or copied (SANS, 2021). Some organizations found that not only were their internal files compromised, but their trusted authentication mechanisms (like single sign-on systems) had been undermined (CISA, 2021). The chilling effect was immediate:

companies and agencies scrambled to assess what was viewed, what was stolen, and how deeply their operations had been infiltrated.

Perhaps more fundamentally, the breach shook faith in digital supply chains and routine IT practices. Organizations had to question whether software updates typically meant to increase security could be trusted at all. “The SolarWinds incident made it clear that you are only as secure as your weakest third-party vendor,” observed one senior cybersecurity advisor (CrowdStrike, 2021). As a result, the attack sparked renewed calls for “zero trust” security architectures, where every user, device, and application is continuously verified and monitored a move that has significant implications for workplace privacy and day-to-day digital life (CISA, 2021).

The U.S. government’s response to SolarWinds was immediate and sweeping. Emergency directives required federal agencies to disconnect affected systems, reset credentials, and launch full-scale incident response operations (CISA, 2021). Congressional hearings, independent reviews, and public–private collaborations followed. The incident prompted a national conversation about the security of the software supply chain, the need for better information sharing, and the persistent challenge of attributing sophisticated attacks to specific state actors without escalation (ENISA, 2022).

At the industry level, vendors and security firms reassessed their own development and update processes. Enhanced code-signing procedures, mandatory security audits, and new standards for software provenance became more common in the wake of the attack (SANS, 2021).

The SolarWinds case stands as a cautionary tale about the interconnectedness and fragility of the digital ecosystem. It demonstrated that even the most security-conscious organizations are vulnerable if the tools they depend on are compromised at the source. It also forced a reckoning with uncomfortable truths about privacy: that sensitive data can be quietly surveilled and stolen for months without anyone noticing, and that trust in basic technology can be shattered overnight.

For many, the most lasting legacy of SolarWinds is the realization that cybersecurity must be a shared responsibility across governments, industries, and even ordinary users. It is a reminder that privacy and security are not just technical problems, but collective societal challenges that will require constant vigilance, adaptation, and trust-building in the years ahead (CISA, 2021; ENISA, 2022).

4.1.3 COLONIAL PIPELINE RANSOMWARE ATTACK (2021)

Just The Colonial Pipeline ransomware attack of May 2021 marked a turning point in public awareness of how cybercrime sometimes state-tolerated or even state-enabled can spill over into everyday life and national security. While it was not the first major ransomware incident, the scale and consequences of this attack made it a global headline and led to real-world disruptions that affected millions of people (CISA, 2021; BBC, 2021).

Colonial Pipeline is the largest refined oil pipeline in the United States, delivering nearly half of the East Coast's fuel supply. In early May 2021, the company was forced to shut down its operations after its business network was compromised by a Russia-based cybercriminal group known as DarkSide (CISA, 2021). The attackers infiltrated Colonial's systems, encrypted their data, and demanded a multi-million dollar ransom in cryptocurrency for its release.

What set this attack apart was the attackers' strategy: they deliberately targeted the corporate IT network, but the shutdown of business operations forced the company to halt pipeline operations as a precaution. The resulting disruption caused widespread fuel shortages, panic buying, and long lines at gas stations across multiple U.S. states (BBC, 2021). This illustrated how a breach of digital systems can have immediate, tangible effects on physical infrastructure and public well-being.

Although DarkSide was a criminal group rather than an official government agency, the fact that it operated with relative impunity from within Russia where ransomware gangs are often allowed to function so long as they avoid domestic targets raised important questions about state responsibility (ENISA, 2022). Some cybersecurity

experts argued that the incident was a form of “state-enabled” cyber aggression, blurring the line between traditional crime and state-backed warfare (Dragos, 2021).

From a privacy standpoint, the Colonial Pipeline incident did not involve mass leaks of personal data. Instead, its lessons for privacy were more indirect but deeply relevant. First, the attack showcased how ransomware actors increasingly target organizations whose disruption will have cascading effects on millions of ordinary people essentially holding a region’s critical services hostage (CISA, 2021).

Second, the aftermath of the attack led to a surge in data collection, monitoring, and law enforcement interventions, including U.S. government efforts to trace and recover ransom payments. Some privacy advocates worried that the urgent response to such crises could open the door to expanded surveillance or regulatory overreach (ENISA, 2022). For example, the U.S. Department of Justice was able to track and seize a portion of the ransom paid by Colonial, thanks in part to blockchain analytics demonstrating both the power and privacy risks of new investigative techniques (BBC, 2021).

The attack was a wake-up call for both policymakers and industry leaders. President Biden declared a state of emergency, and the Department of Homeland Security issued new cybersecurity mandates for pipeline operators (CISA, 2021). The U.S. government began pushing for stronger regulations on critical infrastructure, better reporting of ransomware attacks, and enhanced public-private cooperation. Cyber insurance companies, meanwhile, started reevaluating their coverage terms, and organizations of all kinds invested more heavily in backup systems, network segmentation, and employee training (ENISA, 2022).

One notable outcome was a renewed debate over whether to ban ransom payments altogether a complex issue, since refusing to pay can put victims in impossible positions, while paying incentivizes further attacks.

The Colonial Pipeline incident is now a case study in how cyberattacks can cross over from the digital world into the physical one, with consequences for entire economies

and societies. It demonstrated that resilience is not just about firewalls and antivirus software, but about coordinated action, transparency, and a whole-of-society approach to cybersecurity (CISA, 2021; ENISA, 2022).

Perhaps most importantly, it reinforced the lesson that privacy and security are tightly linked. Disruptions to essential services can lead to secondary privacy impacts, from data monitoring in crisis response to the exposure of sensitive operational information. In an era of growing digital interdependence, defending one means defending the other.

4.1.4 U.S. ELECTION INTERFERENCE (2016 & 2020)

The cyber-enabled interference in the U.S. elections of 2016 and 2020 stands as a dramatic illustration of how digital threats can undermine trust in democratic processes. Unlike classic cyberwarfare, which targets infrastructure or military systems, election interference is aimed directly at the foundations of civil society: public discourse, voter confidence, and the legitimacy of outcomes (ODNI, 2021; Mueller, 2019).

The 2016 U.S. presidential election was a watershed moment for the weaponization of cyber capabilities in the political arena. Russian state actors specifically the GRU and the Internet Research Agency conducted a multifaceted campaign that combined hacking with coordinated online disinformation. Key events included the spear-phishing and subsequent hacking of the Democratic National Committee (DNC) and John Podesta's emails, followed by the strategic release of these communications to embarrass and distract opponents (Mueller, 2019; DHS, 2017). These operations were timed to maximize their influence on public opinion and disrupt the news cycle.

At the same time, Russian operatives created thousands of fake social media accounts to amplify divisive narratives, sow confusion, and erode trust in the election process. Studies show that millions of Americans were exposed to such content, which targeted political groups on both the left and right (ODNI, 2021).

By the time of the 2020 election, both the U.S. government and its adversaries had learned from 2016. The federal government established new bodies like the Cybersecurity and Infrastructure Security Agency (CISA) to defend electoral infrastructure and improve public awareness (CISA, 2020). Meanwhile, foreign actors including not just Russia, but also China and Iran pursued influence operations, attempting to hack campaigns, spread fake news, and even threaten individual voters through coordinated phishing and email spoofing (ODNI, 2021).

Unlike 2016, the 2020 election saw far fewer successful breaches of core election infrastructure. Most attempts focused on perception using social media, deepfakes, and targeted emails to raise doubts about mail-in ballots, polling security, or even the legitimacy of the entire process (ODNI, 2021). One widely cited example was the “Proud Boys” email campaign, which used spoofed addresses to intimidate voters and spread confusion in the final weeks before the election (CISA, 2020).

Both election cycles exposed vast amounts of personal and sensitive information. The DNC and Podesta hacks not only revealed campaign strategies but also private conversations, donor lists, and personal data of staff and supporters (Mueller, 2019). The widespread spear-phishing in 2020 demonstrated how even ordinary citizens could be caught up in global cyber battles.

More broadly, the need to protect electoral systems led to increased monitoring of digital platforms, campaign activity, and social media content. This surge in surveillance and content moderation though often necessary to prevent manipulation sparked debates about censorship, privacy, and the limits of state intervention (ODNI, 2021; ENISA, 2022).

In response to these threats, U.S. lawmakers pushed for new rules governing transparency in political ads, mandatory disclosure of foreign influence attempts, and expanded funding for cybersecurity in elections (CISA, 2020). Tech companies were pressured to better police their platforms, label disinformation, and cooperate with authorities, but many critics argue that such efforts still fall short of the challenge (ENISA, 2022).

Internationally, the U.S. experience became a case study for other democracies. Countries in Europe and beyond began adopting “election resilience” frameworks, integrating cybersecurity with traditional election integrity measures, and preparing for future campaigns marked by hybrid threats (ODNI, 2021).

Perhaps the most lasting effect of the U.S. election interference campaigns has been the damage to trust both in institutions and in the information ecosystem itself. Voters now confront a flood of conflicting claims, “fake news,” and suspicion about the privacy of their own votes and data. The sense that elections can be manipulated from afar has changed how people engage with democracy itself (Mueller, 2019).

At the same time, greater public awareness has prompted citizens to be more cautious, more media-literate, and in some cases, more engaged in defending their own digital privacy. The challenge remains ongoing, but the lesson is clear: the battle for the ballot box is now as much digital as it is physical.

4.1.5 BYBIT \$1.5B CRYPTO HACK (2025)

The The Bybit \$1.5 billion crypto hack, projected here as a plausible near-future event, encapsulates the evolving relationship between state-backed cyberwarfare, financial systems, and personal privacy in the digital era. While Bybit itself is a real global cryptocurrency exchange, the scenario described draws from actual trends and major incidents in digital asset security up to 2025 (Chainalysis, 2025; Kaspersky, 2025).

In this scenario, a sophisticated group widely believed to have state backing or at least state tolerance exploited zero-day vulnerabilities in Bybit’s transaction platform. The attack was multifaceted: it included highly targeted spear-phishing against key employees, supply-chain compromise of software updates, and the deployment of advanced persistent malware to maintain stealthy access over several months (Kaspersky, 2025; ENISA, 2025).

When the heist finally unfolded, over \$1.5 billion in various cryptocurrencies were siphoned from “hot wallets” across millions of user accounts. The attackers immediately leveraged mixers, decentralized exchanges, and cross-chain “bridges” to

launder the stolen funds, making the financial trail extremely difficult to follow (Chainalysis, 2025). Within days, news broke globally, shaking investor confidence and triggering panic withdrawals on other platforms.

While the sheer amount of money grabbed was staggering, the true impact went beyond financial loss. In the process, the attackers gained access to a treasure trove of user data: names, email addresses, phone numbers, KYC documentation, transaction histories, and wallet addresses (Kaspersky, 2025). For millions of users, this breach didn't just mean a loss of funds, but a violation of personal privacy that could put them at risk of future scams, identity theft, or government targeting in restrictive jurisdictions.

As news spread, Bybit and regulatory agencies raced to assure users, freezing remaining assets and launching forensic investigations. Blockchain analytics companies were enlisted to trace the flow of stolen coins—an effort that, while innovative, raised new concerns about user surveillance and privacy on supposedly “decentralized” platforms (Chainalysis, 2025).

Governments worldwide moved quickly. Some, citing national security and anti-money laundering concerns, imposed emergency regulations on crypto exchanges, mandating stronger user verification and transaction reporting (ENISA, 2025). Others called for global treaties on digital asset security and even discussed creating “kill switches” for suspect transactions—a controversial idea that pitted the need for collective safety against the ideals of privacy and autonomy at the core of the crypto movement.

The hack also reignited debates about the responsibilities of exchanges versus users in securing digital assets. Critics argued that even well-funded platforms like Bybit had not invested enough in regular code audits, third-party penetration testing, or multi-signature cold storage for user funds (Kaspersky, 2025). Meanwhile, ordinary users, many of whom had flocked to crypto for its promise of privacy and control, found themselves exposed and unprotected.

At a human level, the Bybit hack triggered waves of anxiety, not just among direct victims but across the global crypto community. Online forums filled with stories of lost life savings, while others shared tips on securing wallets and avoiding phishing schemes. Activists and journalists in repressive countries, who had relied on crypto for donations and communication, worried that their leaked identities could now expose them to state surveillance or retribution (ENISA, 2025).

At the same time, the event showcased the resilience and adaptability of the crypto space. White-hat hackers, blockchain forensic experts, and grassroots advocacy groups collaborated to track stolen assets, pressure exchanges to block tainted funds, and educate the public about privacy best practices (Chainalysis, 2025).

The Bybit hack demonstrated that digital finance is both a target and a tool in modern cyberwarfare. State actors see economic disruption, privacy violations, and loss of trust as weapons, while ordinary people increasingly depend on digital platforms for freedom and opportunity. As crypto adoption accelerates, so too must efforts to strengthen security, defend privacy, and build international standards that protect individuals and societies alike (ENISA, 2025; Kaspersky, 2025).

This case stands as a warning: in the age of decentralized money, privacy and security are not just technical issues but global public goods that demand constant attention from users, platforms, and policymakers alike.

4.1.6 ETHIOPIA–EGYPT CYBER ATTACKS ON THE GRAND ETHIOPIAN RENAISSANCE DAM (GERD) – 2020

The Grand Ethiopian Renaissance Dam (GERD) has stood at the centre of escalating geopolitical tension between Ethiopia and downstream nations, particularly Egypt. In 2020, as diplomatic negotiations over Nile water rights deteriorated, Ethiopia experienced a wave of cyberattacks traced to pro-Egyptian hacktivist groups, with indications that some may have had state backing. These operations targeted government websites, circulated disinformation, and allegedly aimed to compromise digital systems associated with the dam project.

Strategically, the attacks marked a turning point in regional hostilities introducing cyberwarfare as a viable tool for applying pressure without resorting to military confrontation. At a time when direct conflict would have drawn international scrutiny, cyber operations enabled adversaries to disrupt and delay with plausible deniability. Ethiopian government portals were temporarily taken offline or defaced, and a coordinated misinformation effort on social media attempted to generate public alarm about the dam's operational safety.

Though no physical infrastructure was damaged, the psychological and political ramifications were significant. These attacks highlighted how major infrastructure initiatives in Africa especially those with transboundary implications can become flashpoints in the cyber domain. The dam's symbolic and strategic value made it a prime target, demonstrating that water security disputes can now escalate into digital theatres of conflict.

From a privacy and information integrity perspective, the incident did not involve mass data theft but raised concerns over the weaponization of information. The manipulation of public narratives through disinformation blurred the line between strategic communication and psychological operations, directly impacting civilian trust in official government channels. The episode exposed the fragility of digital sovereignty in emerging economies, where state messaging can be quickly undermined by well-coordinated cyber campaigns.

4.2 CROSS-CASE DISCUSSION

Across these diverse but interconnected incidents, a few critical patterns emerge:

- I. **The expanding attack surface:** Nation-state and proxy attackers now target everything from critical infrastructure and elections to private industry and financial markets, increasing the odds that bystanders (both nations and individuals) will become collateral victims (ENISA, 2022; CISA, 2021).

- II. **Blurring lines:** The distinction between espionage, war, crime, and activism is less clear than ever. Ransomware gangs and hacktivists often operate in the same digital space as state actors, and sometimes serve as tools for plausible deniability (CISA, 2021).
- III. **Privacy as a battlefield:** Personal and organizational privacy is not simply a bystander casualty—it is now a direct target and sometimes even a weapon (as in the case of doxxing or data leaks designed to intimidate) (ODNI, 2021; DHS, 2017).
- IV. **Legal and policy lag:** The legal frameworks and international agreements intended to protect privacy and establish norms of cyber conduct have consistently struggled to keep pace with technological change (ENISA, 2022; Mueller, 2019).
- V. **Resilience and surveillance trade-offs:** States and private organizations alike have invested in more robust security practices, but these often involve heightened monitoring, extensive data collection, and even new forms of algorithmic surveillance (CISA, 2021; ENISA, 2022). The drive for security can sometimes come at the cost of civil liberties and personal privacy (ODNI, 2021).

4.3 HUMAN PERSPECTIVE AND PRACTICAL IMPACT

What ultimately makes these cases resonate is not just their technical or geopolitical intrigue, but their very real consequences for ordinary people. Whether it's the blackout in Ukraine, the leak of sensitive emails, the shortage at the fuel pump, or the fear of personal financial loss after a major crypto hack, these incidents shape public trust in both digital and government institutions (Greenberg, 2018; ENISA, 2022). Citizens everywhere now live with the possibility that their data, privacy, and security may be compromised not by accident or negligence, but as part of a deliberate strategy in global power struggles (CISA, 2021).

At the same time, the last decade has shown that resilience is possible. Ukraine's ability to maintain government continuity and public services during hybrid attacks, the U.S. response to election interference, and global cooperation after SolarWinds demonstrate that effective defence and adaptation can blunt even the most serious digital assaults though always at a price (CISA, 2022; ENISA, 2022).

4.4 SUMMARY OF FINDINGS

The analysis of major cyberwarfare events from 2015 to 2025 reveals a decade in which the digital domain truly became a global battleground, shaping not just military strategy but also economics, politics, and the daily lives of millions. Each case study Ukraine–Russia cyberwar, SolarWinds hack, Colonial Pipeline ransomware attack, U.S. election interference, Bybit \$1.5B crypto hack, and the Ethiopia–Egypt cyberattacks on the Grand Ethiopian Renaissance Dam (GERD) in 2020 sheds light on the evolving techniques and far-reaching impacts of cyber conflict.

These events demonstrate that cyberwarfare is no longer the exclusive domain of military or intelligence agencies. Instead, it is carried out by an ever-expanding array of state and non-state actors, including criminal groups, proxies, and politically motivated hackers (ENISA, 2022; CISA, 2021). What ties these incidents together is their ability to cross borders effortlessly, target critical infrastructure, and produce consequences that ripple far beyond the original target.

For instance, the Ethiopia–Egypt GERD cyber conflict, while less publicized than some Western cases, illustrates how digital attacks are now tools of leverage in regional disputes over vital resources. In 2020, as tensions mounted over the filling of the dam, both countries were reportedly targeted by cyber campaigns aimed at government networks, infrastructure, and media outlets seeking to steal sensitive negotiations data, disrupt communications, and influence global opinion (Al Jazeera, 2020; FireEye, 2021). This case underscores the new reality that cyberwarfare is being used not just for power projection, but as a means to influence diplomacy, development, and even water security.

Looking across all these cases, several patterns stand out:

- a. **The attack surface is wider and more unpredictable.** No sector is immune, from elections and oil pipelines to cloud platforms and even the physical flow of water in Africa.
- b. **Privacy is increasingly at risk.** Attacks are not limited to stealing state secrets they can expose the personal data of millions, threaten activists and journalists, and erode public trust in digital systems (ODNI, 2021; Greenberg, 2022).
- c. **Attribution and response remain difficult.** Even when a state is widely believed to be responsible, diplomatic, legal, and technical hurdles often delay accountability, while victims scramble to recover and adapt.
- d. **Legal and policy frameworks lag behind.** Governments and international bodies struggle to keep pace with the rapid evolution of threats, sometimes trading privacy for security, or adopting surveillance tools that may outlast the crisis itself (ENISA, 2022).
- e. **Resilience is possible, but costly.** As seen in Ukraine's cyber defense and the global reaction to SolarWinds and Colonial Pipeline, rapid collaboration and information sharing can blunt attacks, but always require vigilance and investment.
- f. **The human factor is central.** Whether it's an election disinformation campaign, a fuel shortage, a wiped-out crypto wallet, or a dam caught in regional power struggles, the true impact of cyberwarfare is measured in disrupted lives, lost privacy, and shaken trust.

As this decade draws to a close, these case studies remind us that cyberwarfare is now an unavoidable reality of global affairs one that calls for not only technological solutions but also new forms of international cooperation, legal innovation, and above

all, a renewed commitment to protecting human dignity and privacy in a connected world.

CHAPTER 5 CONCLUSIONS AND RECOMMENDATIONS

As cyberwarfare continues to advance, so too must our grasp of its profound and expanding consequences. This chapter weaves together the study's central insights, offering a synthesized reflection on the strategic, ethical, and human dimensions of digital conflict. It closes by outlining actionable recommendations across policy, legal, and research domains each informed by real-world cyber incidents. By distilling these findings, the chapter aims to provide a forward-looking guide for managing the realities of digital warfare while protecting both national security and individual freedoms.

5.1 SUMMARY OF KEY FINDINGS

This research has shown that cyberwarfare has fundamentally transformed the nature of modern conflict and the fabric of national and personal security. Across all major incidents examined including the Ukraine–Russia cyberwar, SolarWinds hack, Colonial Pipeline ransomware, U.S. election interference, the Bybit \$1.5B hack, and the Ethiopia–Egypt cyber operations against the Grand Ethiopian Renaissance Dam (GERD) the impacts extend well beyond military targets. They disrupt critical infrastructure, destabilize economies, undermine democratic processes, and result in widespread privacy violations.

Key findings include:

- I. **Nation-state cyber operations increasingly target civilian infrastructure and personal data**, making privacy a frontline issue in modern conflict (ENISA, 2022; Greenberg, 2022).
- II. **The attack surface is now global, and the risk of collateral damage is higher than ever.** Disinformation campaigns, ransomware, and supply-chain attacks reach far beyond their original targets, affecting societies at large.
- III. **Legal frameworks and international norms lag behind the pace of technological change**, making it difficult to assign responsibility, deter

aggression, or ensure accountability for privacy breaches (Gervais, 2022; ENISA, 2022).

- IV. **The response to cyberwarfare requires not only technical solutions but also policy innovation, ethical consideration, and new forms of international collaboration.**

5.2 CONCLUSIONS

Cyberwarfare The past decade has proven that cyberwarfare is now an enduring feature of international relations. It is a tool for both conflict and coercion, wielded in pursuit of political, economic, and military objectives. Most troublingly, the erosion of privacy has become a routine side effect and sometimes a deliberate aim of state-backed cyber operations.

Three main conclusions emerge:

1. **The boundaries between war and peace, crime and conflict, civilian and combatant have blurred** in the digital era. The cases studied especially the Ukraine–Russia conflict and the GERD cyber dispute demonstrate that cyber operations can escalate regional tensions and threaten civilian populations in new and deeply personal ways (ENISA, 2022; Al Jazeera, 2020).
2. **Current legal, policy, and technical frameworks are inadequate** to address the ethical, human rights, and strategic risks of cyberwarfare. Attribution, proportionality, and civilian protection are all deeply challenging when attacks occur at machine speed and cross multiple jurisdictions (Gervais, 2022).
3. **Resilience is possible** but requires a shift from purely defensive measures to proactive, adaptive, and cooperative approaches. Ukraine’s cyber defense, cross-industry responses to SolarWinds and Colonial Pipeline, and international monitoring during election interference offer models for whole-of-society resilience.

5.3 SUGGESTIONS FOR FURTHER RESEARCH

Future research should explore:

- a. **Empirical Research:** Future studies should use real-time data collection and direct interviews with cybersecurity professionals, policymakers, and affected communities.
- b. **Expanding the Scope:** Additional case studies from Latin America, Asia, and Africa can provide a more global perspective on how cyberwarfare and privacy deprivation manifest in different political and social contexts.
- c. **Legal and Ethical Frameworks:** Interdisciplinary research should further explore the intersection of international humanitarian law, human rights, and digital sovereignty.

5.4 PRACTICAL IMPLICATIONS

- a. **For Governments:** There is an urgent need to strengthen not just cyber defenses but also the legal and ethical frameworks that govern cyber conflict. This means clarifying rules of engagement, sharing intelligence, and protecting the rights of citizens even during periods of digital crisis.
- b. **For International Organizations:** Coordinated action is needed to develop and enforce global standards on cyberwarfare and privacy protection. Platforms like the United Nations and African Union can serve as conveners for new treaties and response mechanisms.
- c. **For Cybersecurity Professionals and Technology Companies:** The stakes are higher than ever for securing critical infrastructure and protecting user data. Regular security audits, strong encryption, and transparent incident response protocols are no longer optional they are fundamental.

- d. **For Civil Society:** Awareness and advocacy matter. Privacy education, media literacy, and digital rights activism can help citizens protect themselves and hold both governments and corporations accountable.

5.5 LIMITATIONS

This research, while comprehensive in its qualitative analysis, is constrained by the inherent challenges of attribution in cyber operations, rapidly evolving tactics, and limited public information on classified or ongoing incidents. The reliance on secondary data and case studies may also omit emerging threats or future developments in the field.

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